Quantifying Model Uncertainty via Loss Landscape Sharpness in Deep Learning

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Motivation

Deep learning models have shown remarkable performance across diverse tasks, yet their black-box nature raises concerns about their reliability in safety-critical applications. In response, there have been continuous attempts to quantify uncertainty. Various methods, such as *MC Dropout* (Gal & Ghahramani, 2016) or *Deep Ensembles* (Lakshminarayanan et al., 2017) , are proposed to quantify uncertainty by sampling diverse sub-models.

"Is stability under dropout-based parameter sampling alone enough to consider a model truly reliable?"

This question highlights a critical limitation: in the context of trust-worthy AI, merely observing stable predictions under the parameter sampling may be insufficient to consider a model truly reliable. From this perspective, Sharpness-Aware Minimization (SAM) (Pierre Foret et al., 2020) has drawn attention to the connection between generalization performance and the sharpness of the loss landscape. That is, models that lie in flatter minima tend to generalize better. We argue that the reliability of the trained model should be evaluated through not only prediction stability but also loss landscape geometry. Motivated by this idea, our work focuses on the distinction between the model uncertainty measured via MC Dropout and the sharpness of the loss landscape. Specifically, we try to investigate how different sub-model sampling strategies—random dropout versus SAM-style perturbations—affect this relationship, focusing on the differences in their parameter sampling strategies.

Ultimately, we try to confirm the possibility that SAM-style perturbations offer a promising direction for developing new uncertainty quantification methods, grounded not only in predictive variation but also in geometric properties of the model's solution space.

Parameter Sampling Strategy & Quantification Metrics

Trained Parameter

- Parameter ω of the deep learning model $f(\cdot)$ is sampled from $p(\omega|\mathcal{D})$ through

$$\omega = arg\min_{\omega} \mathcal{L}(f(x;\mathcal{D},\omega),y)$$

MC Dropout (dropout-based parameter sampling)

- Yarin Gal et al., 2016 proposed a *MC Dropout*, estimating the predictive distribution by sampling the trained model's parameters with dropout mask, to generate multiple predictions by activating dropout.
- With dropout, binary variable z_{ij} is multiplied for each parameter $\omega_{i,j}$ at the j^{th} node of i^{th} layer.

$$\omega_{i,j}^{t} = z_{ij} \odot \omega_{i,j} \text{ with } z_{ij} \sim Berrnoulli(1-p)\text{, where dropout rate } p \text{ for } t = 1,2,\dots T$$

SAM-based Perturbation

- *SAM* (Pierre Foret et al., 2021) improves generalization by seeking flat minima in the loss landscape.
- *SAM* utilize the perturbation to model's parameter toward the direction of steepest loss increase around ρ:

$$\omega^{*} = \omega + \rho \cdot \frac{\nabla_{\omega}\mathcal{L}(f(x;\omega),y)}{\|\nabla_{\omega}\mathcal{L}(f(x;\omega),y)\|}$$

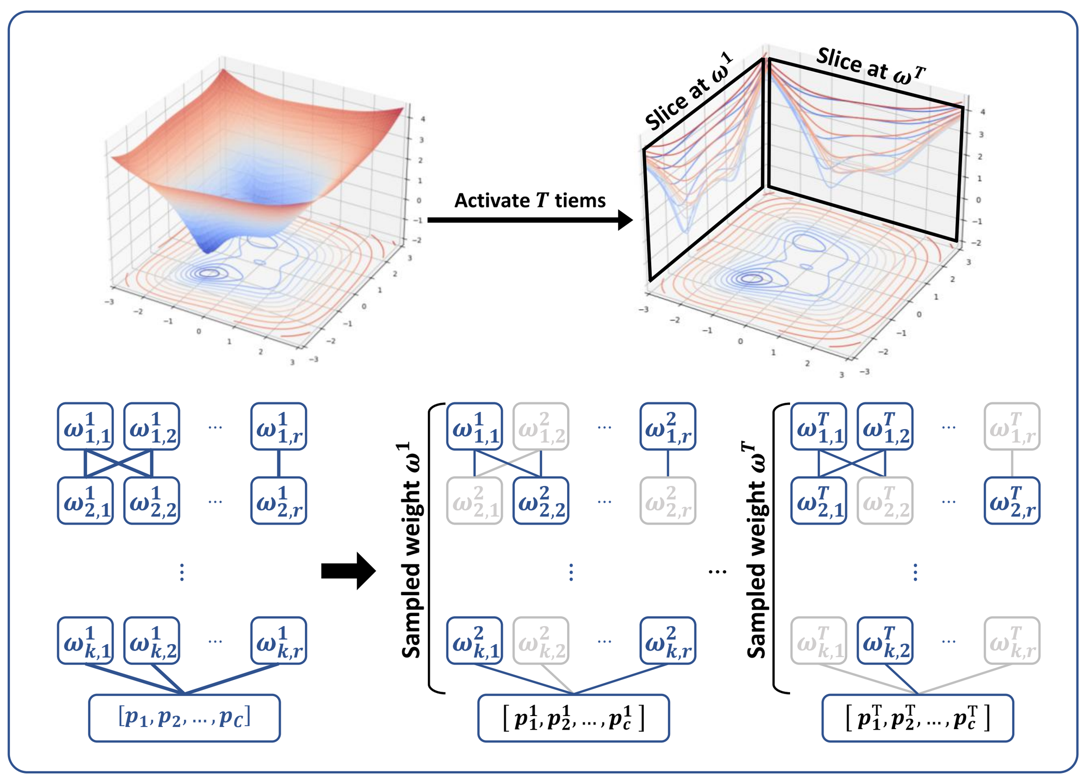

Fig 1. Dropout-based Sampling

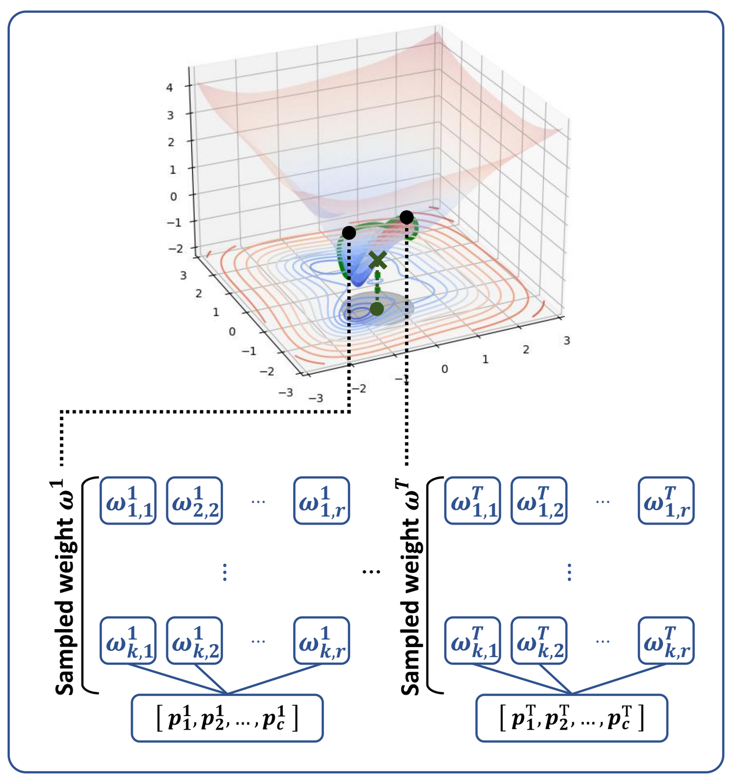

Fig 2. *SAM*-style Perturbation

Dropout-based Sampling Strategy & Metrics

- Predictive variance estimated by T sub model with the sampled parameters $\omega^1, \omega^2, \dots, \omega^T$:

$$Var_T[f(x;\omega^t)] = \mathbb{E}_T[\{f(x;\omega^t) - \mathbb{E}_T[f(x;\omega^t)]\}^2]$$

- At the classification task mutual information(MI), akin to predictive variance, is widely used.

$$MI(x) = \mathcal{H}[{}^{1}/_{T}\textstyle\sum_{t=1}^{T} f(x;\omega^t)] - {}^{1}/_{T}\textstyle\sum_{t=1}^{T}\mathcal{H}[f(x;\omega^t)],$$

$$\text{where } \mathcal{H}(p_1,p_2,\dots,p_C) = -\mathbb{E}_{c=1,2,\dots C}[p_c \cdot \log p_c]$$

- With T sub-models sample by dropout, loss fluctuation also can be measured.

$$\Delta\mathcal{L}_{dropout} = \max_{t} \mathcal{L}(f(x;\omega^t),y) - \mathcal{L}(f(x;\omega),y)$$

SAM-based Sampling Strategy & Metrics

- NS Keskar et al., 2016 used their own sharpness metric, which measures how the loss landscape is sharp with a radius ρ around the parameter ω.

$$\frac{\max_{\|\epsilon_t\|\leq\rho} \mathcal{L}(\omega+\epsilon_t) - \mathcal{L}(\omega)}{1+\mathcal{L}(\omega)} \text{ where } t\text{-th sampled parameter } \omega^t = \omega + \epsilon_t$$

- With T sub-models sample by perturbations within radius ρ around ω, fluctuation of predictions also can be measured.

$$\Delta\,\hat{y}_{SAM} = \max_{t} \|f(x;\omega^t) - f(x;\omega)\|$$

Experiment Ⅰ

Setting

- $T = 10$ sub-models are sampled.
- Used architectures of varying complexity: LeNet5, VGGNet16, WideResNet40-10
- Benchmark dataset CIFAR-10
- Base optimizer with SGD using lr 0.1 and momentum 0.9

Fig 3. 10 Classes of CIFAR-10 dataset

Result 1: LeNet

	Test Error	ECE (↓)	λ_{max}	Sharpness	Model Uncertainty
SAM	1.1973 ± 1.5551	0.3311 ± 0.2798	7.8577e+00	2.0466e-02	0.4252 ± 0.2297
SGD	1.3173 ± 1.7868	0.3366 ± 0.3019	1.3936e+01	3.3769e-02	0.4493 ± 0.2489

- SAM leads to lower Hessian curvature and sharpness score.
- There is no meaningful difference observed with or without SAM.

Result 2: VGGNet

	Test Error	ECE (↓)	λ_{max}	Sharpness	Model Uncertainty
SAM	0.5243 ± 1.3217	0.1517 ± 0.2750	7.2416e+03	2.0727e-04	**0.0821 ± 0.1097**
SGD	0.6839 ± 1.6725	0.1739 ± 0.3022	**1.5148e+04**	**5.1552e-04**	0.1025 ± 0.1358

Result 3: WideResNet

	Test Error	ECE (↓)	λ_{max}	Sharpness	Model Uncertainty
SAM	**0.4181 ± 1.0802**	0.1434 ± 0.2588	1.5249e+05	1.6829e-05	0.0896 ± 0.1087
SGD	0.6839 ± 1.6725	0.1739 ± 0.3022	**3.3019e+06**	**3.2066e-05**	**0.0925 ± 0.1198**

- Models trained with SAM usually show lower average test error, indicating better generalization.
- *SAM* leads to much lower Hessian curvature and sharpness score (over 2 times lower), guiding the model toward flatter minima.
- Sharpness can differ significantly even when model uncertainty measured by *MC Dropout* remains similar, indicating they do not necessarily correlate.
- These patterns are remained not only at the Result 2 but also at the Result 3.

"The result highlights a limitation of dropout-based sampling and motivates the need for **sharpness-aware uncertainty quantification** when evaluating model reliability."

Experiment Ⅱ

Sampling Strategy	Architecture	Fluctuation of Predictions	Fluctuation of Loss
Perturbation	MLP	$8.0426 \cdot 10^{-6}$	$0.0245 \cdot 10^{+0}$
	LeNet5	$8.7006 \cdot 10^{-5}$	$0.0933 \cdot 10^{+0}$
	VGG16	$9.9263 \cdot 10^{-5}$	$0.0856 \cdot 10^{+0}$
	ResNet18	$\mathbf{0.0002 \cdot 10^{+0}}$	$\mathbf{0.2813 \cdot 10^{+0}}$
Dropout	MLP	$0.0177 \cdot 10^{+1}$	$2.4698 \cdot 10^{+1}$
	LeNet5	$\mathbf{0.0226 \cdot 10^{+1}}$	$\mathbf{4.1748 \cdot 10^{+1}}$
	VGG16	$0.0006 \cdot 10^{+1}$	$0.1327 \cdot 10^{+1}$
	ResNet18	$0.0063 \cdot 10^{+1}$	$0.9486 \cdot 10^{+1}$

- Whether we measure fluctuation of predictions or loss, the overall pattern remains consistent across distinctive architectures.
- This highlights the potential of the idea of sharpness as a meaningful basis for uncertainty quantification, beyond existing method *MC Dropout*.

Experiment Ⅲ

Customized Train Set

- Using benchmark dataset MNIST

	Customized Train Set		OoD
Class			
	Seen	Unseen	Unknown

Result

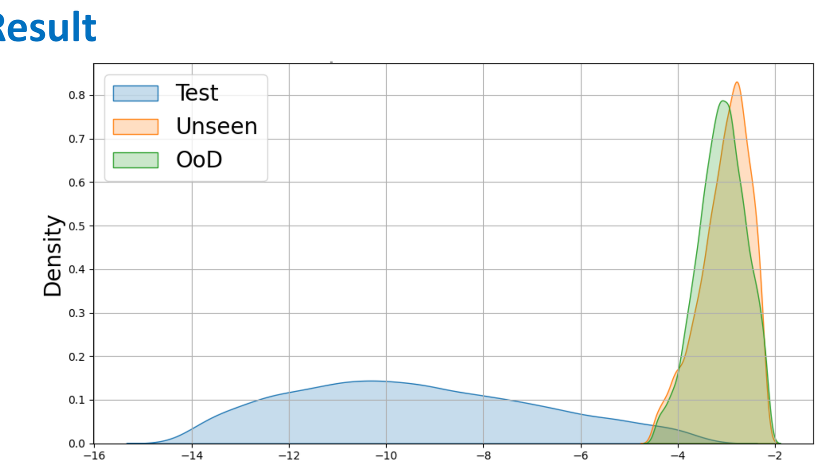

Fig 4. Dropout-based Perturbation

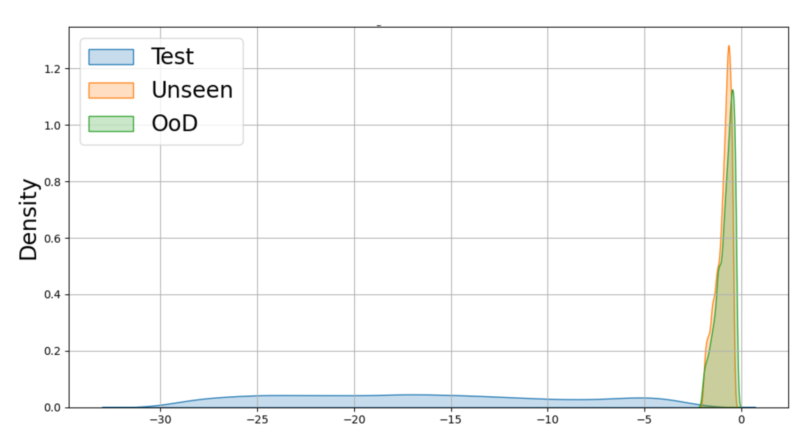

Fig 5. SAM-style Perturbation

Conclusion

- This study reveals that dropout-based sampling, while effective in capturing predictive variability, does not sufficiently reflect the geometric characteristics of the loss landscape, which suggests a potential connection to the development of novel uncertainty quantification methods that better align with the underlying geometry .
- Our analysis shows that relying solely on dropout-based uncertainty may leave out critical aspects related to generalization and robustness.
- Based on this groundwork, we plan to formalize a new uncertainty quantification method that integrates sharpness-aware metrics.

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